PATRICK LIN

Patrick Lin | GitHub | Email | LinkedIn

Computer science student passionate about solving problems with software and artificial intelligence.

EDUCATION

University of Toronto **Toronto, ON**

B.S. in Computer Science *Sept. 2019 - May 2022*

- **GPA:** *3.72/4.00*
- **Coursework**: Introduction to Machine Learning, Software Design, Data Structures & Analysis, Intro. to Databases, Cognitive Psychology, Learning and Memory, Advanced Calculus, Probability & Statistics, Software Tools & Systems Programming
- **Academic Awards:** Dean's List Scholar (Summer 2020), AP National Scholar 2018, PSAT/NMSQT Semifinalist

EXPERIENCE

Informatica *June 2021 – Present*

Software Development Intern

Data Science Toronto *June 2021 – Present*

Project Developer

PROJECTS

Ticker Trackr (https://devpost.com/software/tickertrakr) *February 2021*

*Hackathon Award Winner - **Best Data Visualization** at **Hacklytics** 2021, hosted by **Georgia Tech** Data Science*

- Architected a full-stack web app using **Flask** framework under a 36-hour time limit
- Utilized **Twitter, Reddit, and FinnHub APIs** to retrieve real-time social media data on popular stock tickers

Student Performance Prediction *December 2020*

Trained machine learning models in a team of 2 to predict probability of answering a question correctly

- Implemented Maximum-Likelihood ***Item Response Theory*** model with 70.7% accuracy
- Extended the model to account for more parameters, leading to a decrease in accuracy of 0.1%

University Chat *September 2020 - Present*

Flutter chat app to help university students looking to connect with their course-mates online

- Architected using ***Flutter, Dart, Firebase***, ***Android Studio***, ***Java*** / ***Kotlin***, ***XML***
- Designed Firebase data model for minimal cost, security, and optimal user experience
- Currently implementing student matching using ***Google Cloud functions***

Audio Classification *September 2020*

Neural network model to classify audio samples of kick drums, snare drums, cymbals, claps, hats

- Architected using ***Tensorflow***, ***Jupyter Notebook***, ***Python*** (librosa, scipy, matplotlib)
- Trained a Deep Learning model to determine kick drums vs crash cymbals with 99% accuracy

Music Generation *April 2020 – May 2020*

Automatic generation of lo-fi hip hop beats with jazz & tonal music theory

- Architected using ***Python*** and python libraries, e,g, pyo (python osc.), Mido, PySynth, scipy, librosa
- Implemented composer and processor classes to generate beats in WAV format

Calendar App *Jan. 2020 – April 2020*

Android app developed in a team of 5 for implementation of software design patterns

- Built using ***Java***, ***XML,*** *and* ***Git***
- Learned to apply Software Design patterns, e.g. SOLID principles

SKILLS

Languages: Python (numpy, pandas, matplotlib, scikit-learn, scipy), SQL, R (ggplot2, tidyverse), Java/Kotlin, MIPS Assembly

Frameworks & Tools: Git, LaTeX, HTML/CSS/JS (NodeJS, ReactJS), Android/Flutter

Interests: Cross Country, Tennis, Music Theory/Production/Instruments